Case Study

Half-Year Report (Q3 and Q4 2024) - Atlantis Logistic DOO

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Contents

1. Introduction

This case study provides a comprehensive insight into the operational and financial performance of a local transport company – Atlantis Logistic DOO, focusing on the period of the second half of 2024 (Q3 and Q4).

The goal of the case study is to follow the process from data modeling in the database, through the extraction of raw revenue data, to creating a report that summarizes key performance indicators (KPIs).

The presented set of documents enables an understanding of the business structure and the identification of the main revenue variables (Routes, Clients, Vehicles) during the analyzed period.

For better clarity on individual files within this document, please refer to the files in the main folder.

2. Techniques and Tools Used in This Case Study

- Data Structuring and Modeling - The Foundation
- Data Extraction and Querying - SQL
- Data Processing and Output - Excel CSV
- Reporting and Analysis - Power BI Business Intelligence
  - Key Performance Indicator (KPI) Analysis
  - Revenue Contribution
  - Operational Efficiency

3. Database Schema

As shown in Schema 1, the relevant database sub-schemas used for this case study are highlighted by the red border.

Shema 1 highlights the core components of the relational database model that are used for generating this revenue report.

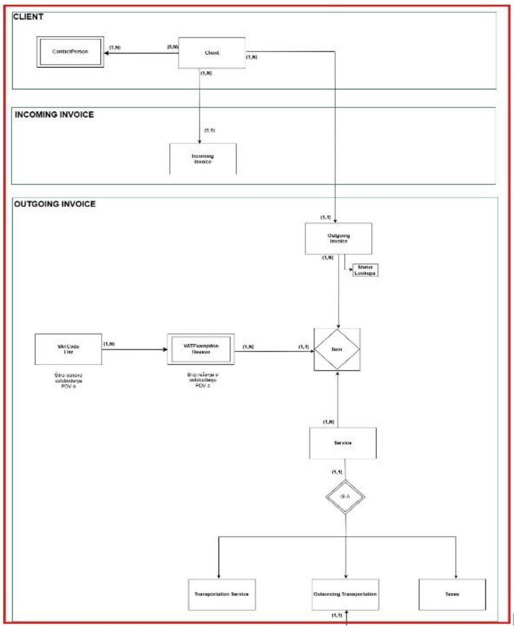
The area, outlined in a red border, represents the logical subsets of the company database that are critical for linking transport operations to financial outcomes.

It establishes the relationship between entities that are the core of this case study:

1. **TransportationService**: Stores details of the service provided, including the Route and the Price (that has been used for calculating the revenue).
2. **Item**: Acts as the link between the service provided and the outgoing invoice. It represents a many-to-many relationship between outgoing invoices and services. (One outgoing invoice can have multiple services and vice versa.)
3. **Outgoing Invoice**: Contains essential invoicing information like the Invoice Number, Issue Date (used to determine Month/Quarter), and the TaxID to identify the client.
4. **Client**: Provides the Client Name linked via the TaxID.

5. **Truck**: Stores the Registration Tag (Reg. Tag) of the vehicle used, linked to the Transportation Service entity.

Eventually, the sub-schemas labeled by the red border are the analytical framework that allows the company to trace a specific transport service back to the client, the vehicle used, and the invoice issued, enabling the calculation and segmentation of the revenue.



Schema 1 – Relevant sub-schemas

4. Database Relational Model

Schema 2 explains the relational model and its key entities, such as **OutgoingInvoice, Item, TransportationService, Client, Truck, Employee** and their primary keys and relationships to other entities (relations). The relation model serves as a blueprint for understanding the data source and how information about transportation services is connected to invoices and revenue.

OutgoingInvoice({InvoiceID, OutInvoiceNmbr, Currency, ReferenceNmbr, OrderNmbr, TransDate, IssueDate, DueDate, Attachment, Note, DocumentStatus, ProcessingStatus, PaymentStatus, TaxID}, {InvoiceID})

Item({InvoiceID, ServiceID, Discount, VATCode, VATExamptionCode}, {InvoiceID, ServiceID})

TransportationService{(ServiceID, Route, Price, TruckID, TrailerID),(ServiceID)}

Truck({TruckID, Make, Model, RegistrationTag,Status}, {TruckID})

Client({TaxID, RegNmbr, ClientName, StreetAndNmbr, City, ZIP, Country, IsActive, Email}, {TaxID})

Schema 2 - Relational Model

5. Revenue Query

Query 1 joins the five most important relations (tables) for this case study: **Transportation Services, Item, Invoices, Clients,** and **Trucks** to export key columns: Invoice Number, Route, Revenue, Vehicle Registration Tag (Reg.Tag), Client name, Month, and Quarter.

```sql
SELECT
      OutInvoiceNmbr,
      Route,
      ts.Price AS Revenue,
      t.RegistrationTag,
      ClientName,
      DATENAME(MONTH, oi.IssueDate) AS Month,
      DATEPART(QUARTER, oi.IssueDate) AS Quarter
FROM TransportationService ts
JOIN Item i ON ts.ServiceID = i.ServiceID
JOIN OutgoingInvoice oi ON i.InvoiceID = oi.InvoiceID
JOIN Client c ON c.TaxID = oi.TaxID
JOIN Truck t ON t.TruckID = ts.TruckID
ORDER BY oi.IssueDate ASC;
```

Query 1 - Revenue SQL Query

6. Revenue Result

Table 1 – Revenue CSV Snapshot

OutInvoice	Route	Revenue	Reg.Tag	Client	Month	Quarter
INV-200	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-13	Budapest (	50000.00	SU600-AE	Masterpla	July	3
INV-201	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-202	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-16	Budapest (	50000.00	SU600-AE	Masterpla	July	3
INV-203	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-19	Budapest (	50000.00	SU600-AE	Masterpla	July	3
INV-204	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-205	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-206	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-207	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-208	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-209	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-210	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-31	Budapest (	50000.00	SU600-AE	Masterpla	July	3
INV-211	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-212	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-213	Olesnica (F	100000.00	NS302-AA	ABM Inter	July	3
INV-37	Budapest (	50000.00	SU600-AE	Masterpla	July	3

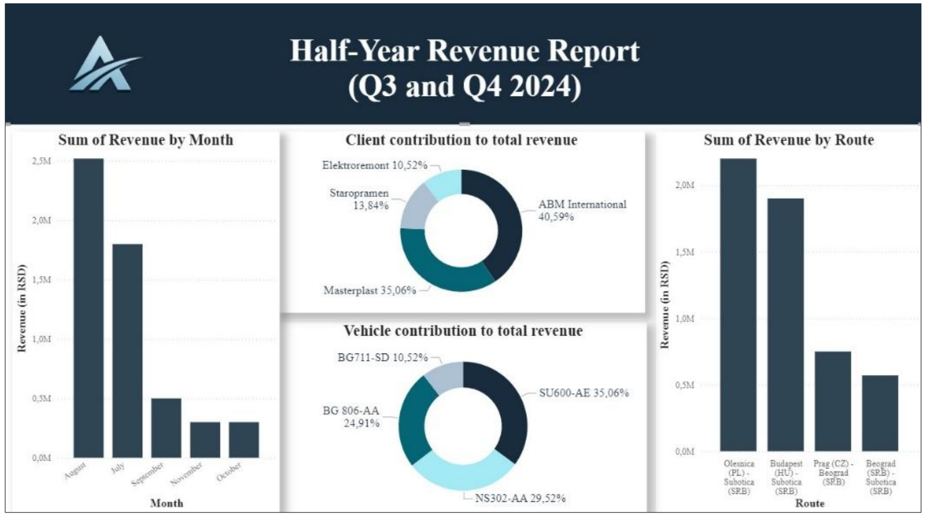
Table 1 displays a CSV snapshot of the SQL query execution. This is the raw revenue data for Q3 and Q4 2024. It serves as the input data for further analysis and reporting in Power BI.

7. Power BI Reports

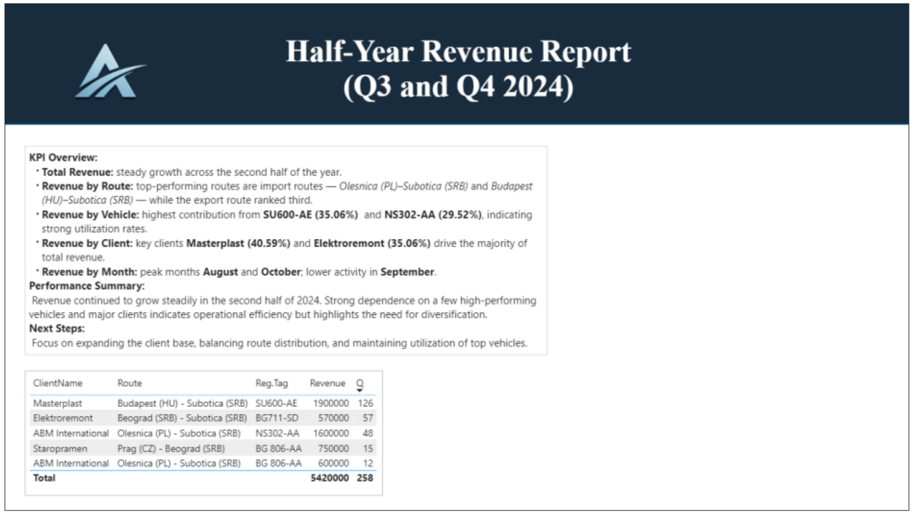
Reports 1 and 2 on Page 6 include an analysis summary that is a KPI Overview with charts. They highlight key insights:

- Steady revenue growth
- Dominant routes (e.g., Olesnica-Subotica)
- Highest vehicle contribution (SU600-AE, NS302-AA)
- Key clients (Masterplast, Elektroremont)
- Conclusion
- Suggestions for improvement in the future

It provides a conclusion on operational efficiency and recommendations (client base diversification).



Report 1 – Charts



KPI Overview:

- **Total Revenue:** steady growth across the second half of the year.
- **Revenue by Route:** top-performing routes are import routes — *Olesnica (PL)–Subotica (SRB)* and *Budapest (HU)–Subotica (SRB)* — while the export route ranked third.
- **Revenue by Vehicle:** highest contribution from **SU600-AE (35.06%)** and **NS302-AA (29.52%)**, indicating strong utilization rates.
- **Revenue by Client:** key clients **Masterplast (40.59%)** and **Elektroremont (35.06%)** drive the majority of total revenue.
- **Revenue by Month:** peak months **August** and **October**; lower activity in **September**.

Performance Summary:

Revenue continued to grow steadily in the second half of 2024. Strong dependence on a few high-performing vehicles and major clients indicates operational efficiency but highlights the need for diversification.

Next Steps:

Focus on expanding the client base, balancing route distribution, and maintaining utilization of top vehicles.

ClientName	Route	Reg.Tag	Revenue	Q
Masterplast	Budapest (HU) - Subotica (SRB)	SU600-AE	1900000	126
Elektroremont	Beograd (SRB) - Subotica (SRB)	BG711-SD	570000	57
ABM International	Olesnica (PL) - Subotica (SRB)	NS302-AA	1600000	48
Staropramen	Prag (CZ) - Beograd (SRB)	BG 806-AA	750000	15
ABM International	Olesnica (PL) - Subotica (SRB)	BG 806-AA	600000	12
Total			**5420000**	**258**

Report 2 - Conclusion

8. Conclusion

This case study executed a complete analytical cycle, from defining the foundational relational database schema and extracting raw data via a precise SQL query to producing a comprehensive Quarterly Revenue Report.

The analysis for the second half of 2024 (Q3 and Q4) confirms steady overall revenue growth, indicating fundamental operational health and efficiency. However, the key insights from the report highlight significant areas of concentration that represent both current strengths and future vulnerabilities:

1. **Route Dominance:** Import routes **Olesnica (PL) –Subotica (SRB)** and **Budapest (HU)–Subotica (SRB)** are the key revenue drivers, demonstrating the company's strong position.
2. **High Client and Vehicle Concentration:** The business exhibits strong dependence on a very small group of entities.
3. **Clients:** Two key clients, **Masterplast (40.59%)** and **Elektroremont (35.06%)**, drive the vast majority of total revenue.
4. **Vehicles:** Two trucks, **SU600-AE (35.06%)** and **NS302-AA (29.52%)**, account for over 64% of revenue, indicating extremely high utilization rates but also a significant operational risk should those specific vehicles face downtime.
5. **Monthly Fluctuation:** The data shows a pattern of peak activity in August and October with a noticeable fall in September, suggesting potential seasonal or scheduling issues that need to be addressed for consistent performance.

Performance Summary

While the company's performance in late 2024 was profitable and efficient, the heavy reliance on specific routes, clients, and vehicles introduces substantial commercial and operational risk.

As outlined in the "Next Steps," the strategic focus must shift towards diversification and risk mitigation:

1. **Client Base Expansion:** Actively pursuing new contracts to reduce the dependency on the two major clients.
2. **Route Balancing:** Developing and promoting other, potentially more profitable, routes (particularly export) to ensure a more balanced revenue distribution.
3. **Fleet Management:** Maintaining the high utilization of top-performing vehicles while simultaneously investing in or ensuring better use of the rest of the fleet to distribute the revenue burden and mitigate the risk of vehicle failure.

In conclusion, the study validates the company’s current operational effectiveness but clearly defines the urgent strategic necessity for broadening its commercial foundations to ensure long-term stability and sustained growth.